

Development of Automated Mobile Robotic Arm for Cold Storage Operations



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Project Guide

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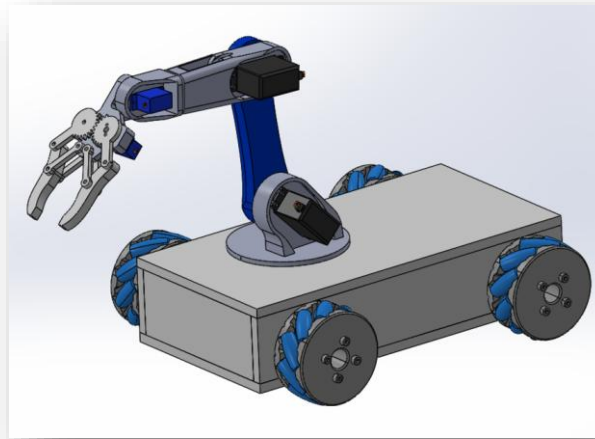
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Literature review



Relevant Technologies:

- Robotics and automation are already widely used in industries like logistics and cold storage. However, most of these systems are either stationary or have limited mobility. Technologies like automated conveyor belts, robotic forklifts, and stationary robotic arms are commonly used to improve efficiency in cold environments.

Gaps in Knowledge:

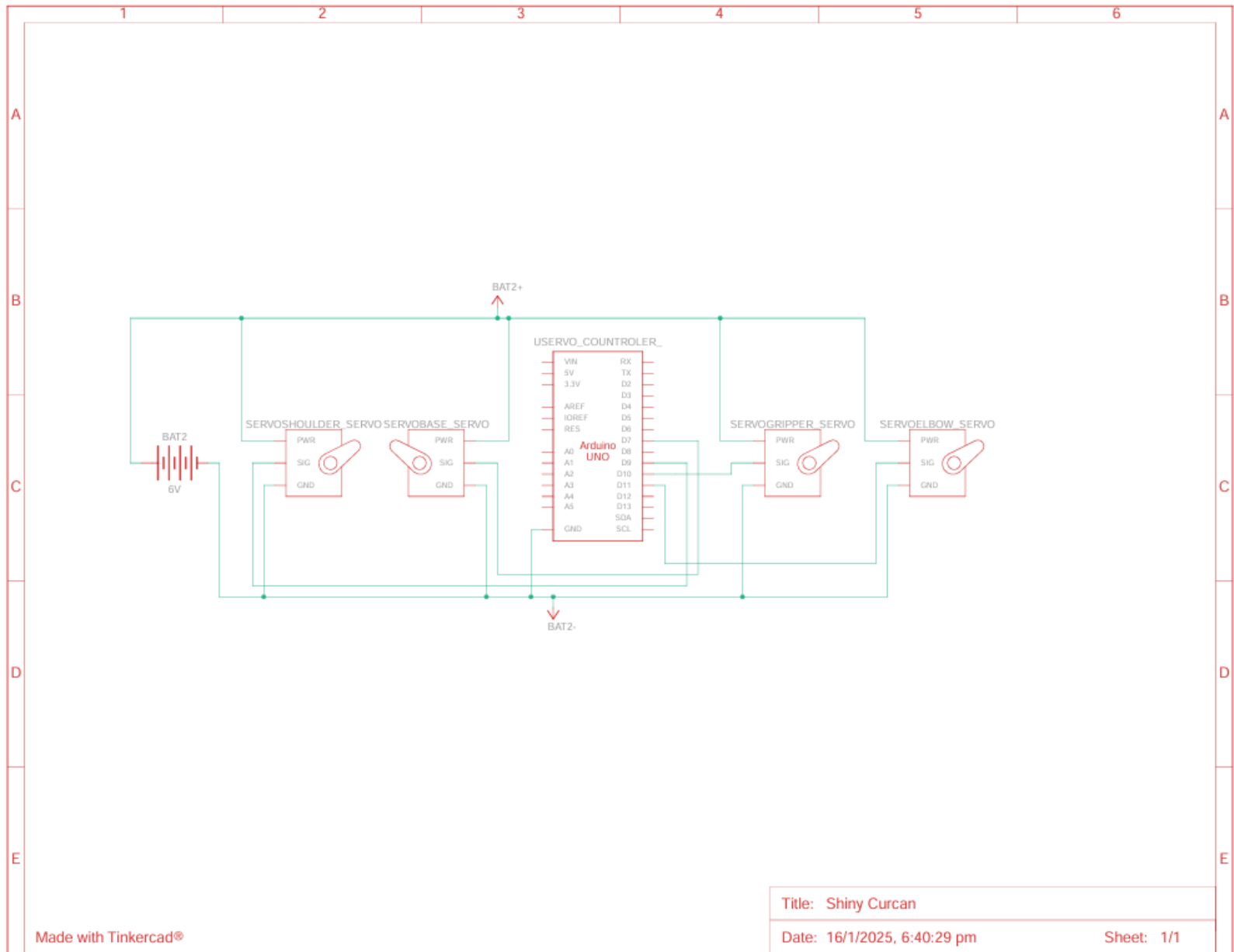
- Although many studies focus on robots working in challenging environments, there is limited research on mobile robotic systems specifically designed to operate in cold storage. Most robotic systems are not built to handle the combination of mobility, precision, and durability required in such low-temperature environments.

Project Setup

Components:

- Servo Motors
- Servo Controller
- Battery
- EEZYbotARM MK2 (3D printed)
- Stepper Motor
- DRV8825 Stepper Motor Driver Module (High Current)
- ControllersTech HC-05 master-slave 6pin Bluetooth Transceiver Module





MECANUM WHEELS
ROBOT CAR

BACK LEFT MOTOR

FRONT LEFT MOTOR

18650 battery

18650 battery

18650 3.7V Li-Ion
TWO BATTERIES

L298N MOTOR DRIVER
02

L298N MOTOR DRIVER
01

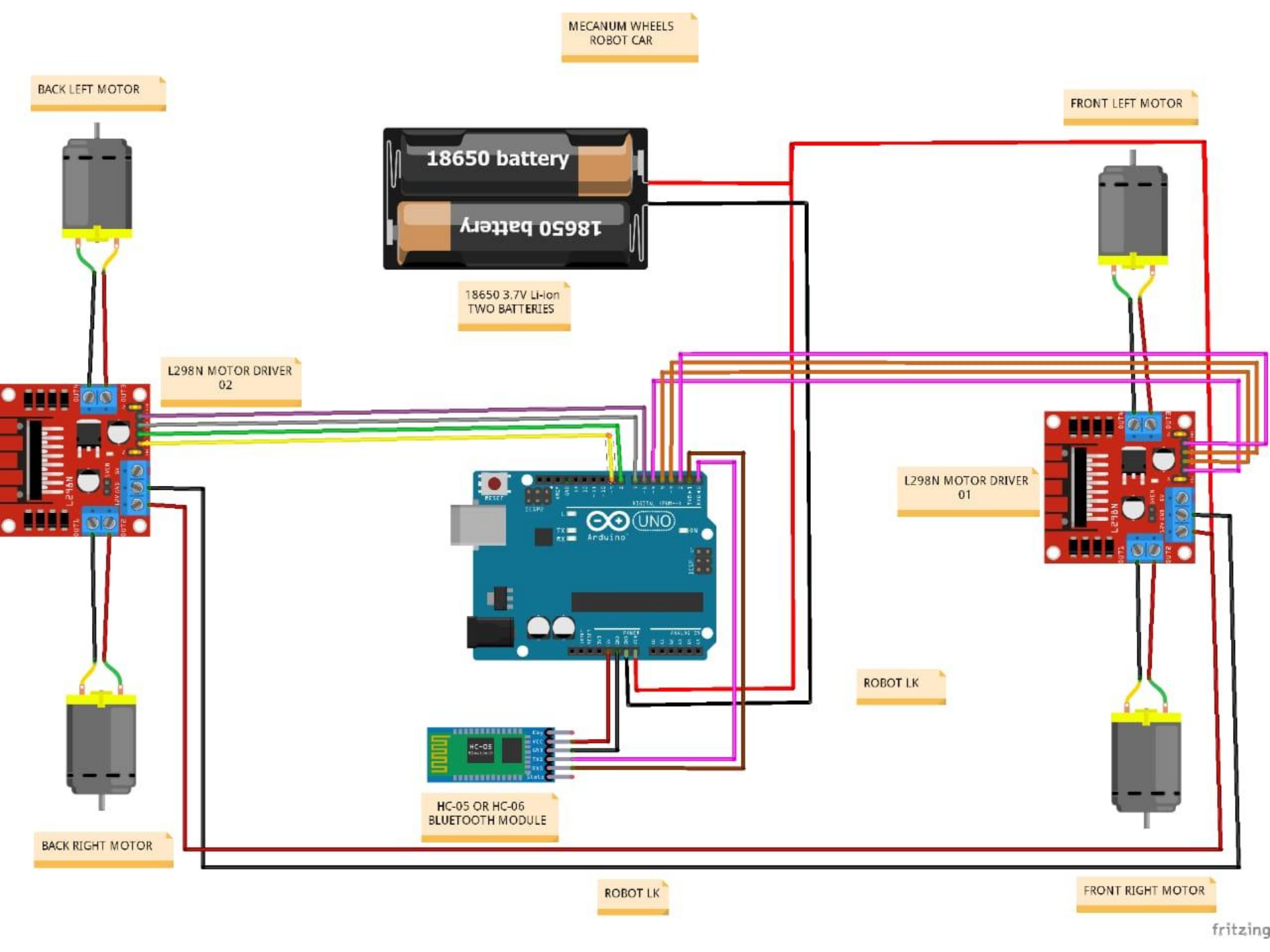
ROBOT LK

HC-05 OR HC-06
BLUETOOTH MODULE

BACK RIGHT MOTOR

FRONT RIGHT MOTOR

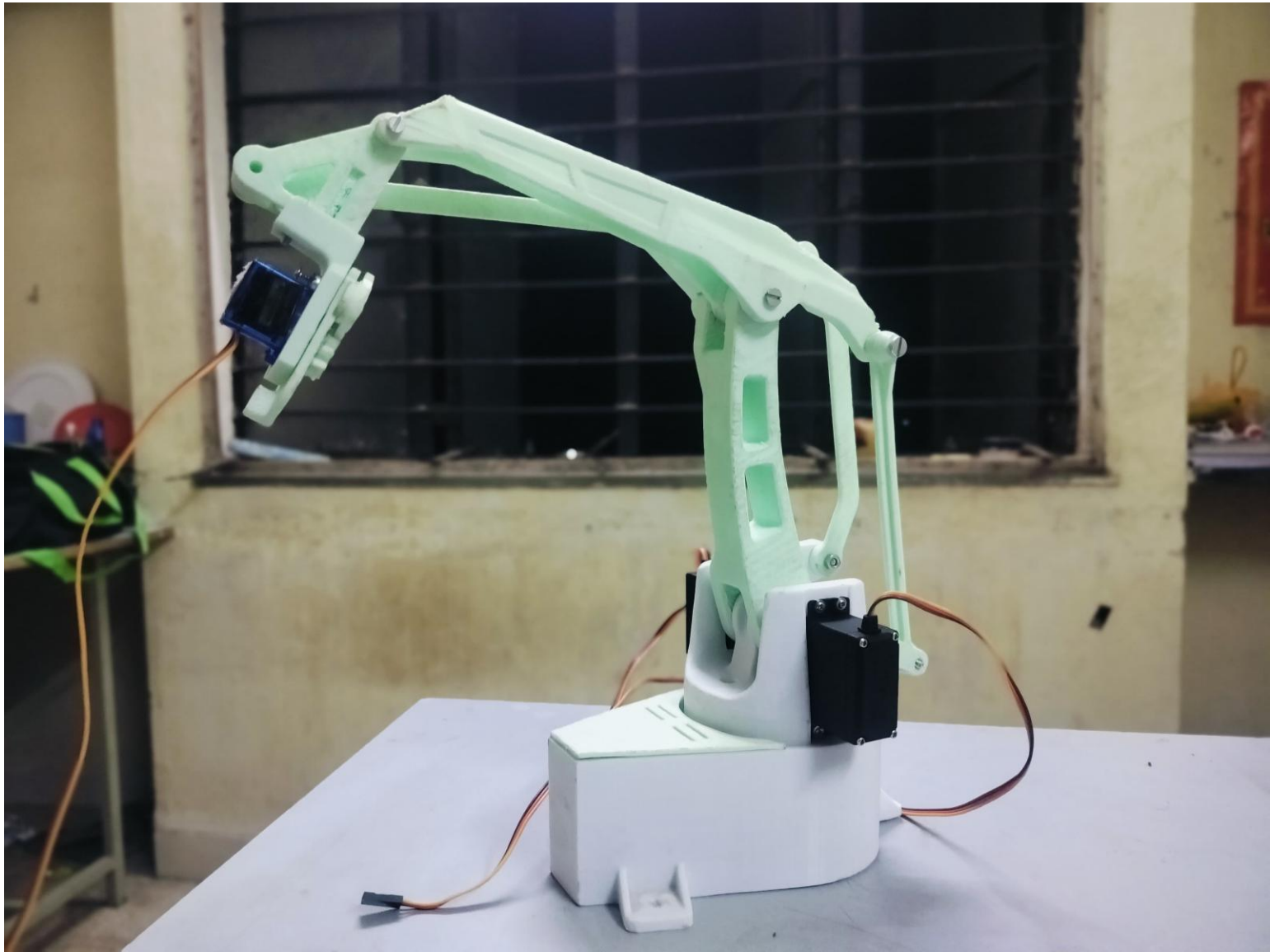
ROBOT LK



Progress of project work

Accomplished Work:

- 3D printed all parts for the project.
- Completed 90-95% of the assembly; remaining assembly is on hold due to pending screw sizes.
- Fully assembled the mobile vehicle with Mecanum wheels.



Robotic arm



Mobile vehicle with Mecanum wheels

Calculations for reference

- When designing an automated mobile robotic arm for cold storage operations, safety is a critical consideration.
- The calculations needed to ensure safety are multifaceted and include structural integrity, overload protection, motion limits, environmental resilience, and fail-safe mechanisms.
- Below are the key safety-related calculations required for the robotic arm design

1. Payload Capacity and Overload Protection

- The arm must be able to handle expected loads safely without risking structural failure or excessive strain on its components.
- The calculation of the payload capacity is based on the weight of the objects the arm will handle, as well as a safety margin to ensure the arm remains within its operational limits.
- Payload Calculation:

Payload (P): This is the maximum weight the robotic arm is designed to lift

$$P_{\text{safe}} = P_{\text{expected}} \times \text{Safety Factor}$$

$$P_{\text{safe}} = 100\text{mg} \times 1.5 = 150\text{mg}$$

This means the arm's components should be rated to handle at least 150mg to account for unexpected stresses.

2. Torque Calculation for Arm Joints

- Each joint in the robotic arm needs to be able to handle the torque required to lift and move the object. The torque exerted by a motor is determined by the load at the end of the arm and the distance from the center of rotation (i.e., the arm's length).

Torque at Each Joint : Torque (T) is calculated as the product of the force exerted by the load (F) and the distance from the pivot point (radius r):

$$T = F \times r$$

For a 200 mg object with a 1-meter arm: $F = 0.200 \times 9.81$

$$F = 1.962 \text{ N}$$

$$T = 1.962 \times 50 = 0.981 \text{ N-m}$$

$$T = 10 \text{ kg/cm}$$

3. Motor Power Calculation

- To calculate the power needed by the motor to operate the robotic arm safely, we need to ensure that the motor can handle the required torque and speed. Motor power (P_{motor}) is given by the formula:

$$P_{\text{motor}} = T \cdot \omega / 1000$$

$$P_{\text{motor}} = 0.981 \cdot 2.05 / 1000$$

$$P_{\text{motor}} = 2.05 \text{ W}$$

Environmental Resilience and Cold-Storage Adaptation

- The components of the robotic arm need to be adapted for low temperatures and humidity levels found in cold storage environments.
- Temperature Resistance of Components:
 1. servo motors (should be rated for low temperatures, typically down to 0°C to -4°C
 2. Lubricants used in the joints must be suitable for low temperatures to prevent freezing or excessive thickening.
 3. Batteries should be rated for cold temperatures, as standard batteries lose efficiency in cold environments. For example, lithium-ion batteries operate less efficiently below 0°C.

Cold Storage Testing

- Now that assembly is almost complete, the next step is testing in a cold storage environment.
- We will evaluate the robotic arm's performance under low temperatures.
- Adjustments and improvements will be made based on test results.

Conclusion

- Successfully assembled the **EEZYbotARM MK2** and **Mecanum Wheel vehicle**.
- Completed **90-95% assembly**, pending minor adjustments due to screw sizes.
- Ready for **cold storage testing** to evaluate performance in low temperatures.
- Future work includes **fine-tuning based on test results** and potential improvements.

THANK YOU